A New Register File Structure for the High-Speed Microprocessor

HIROSHI KADOTA, SUMIO OZAWA, KATSURA KAWAKAMI, AND EISUKE ICHINOHE

Abstract—**The pipeline operations of the register file in a microprocessor are analyzed in detail. Conventional register files, two-port static RAM's, have two problems in successive write-to-read operations. 1) A read-time error takes place when the transition of the *W/R* mode and the transition of register address occur simultaneously. 2) A write-time error takes place when the supply voltage is low. A new register file structure is proposed, which has three address word lines and four data bit lines for each memory cell. This structure enables the independent write and read operations to each other, and can solve the two problems. By using this register file structure, a new 16 bit microprocessor with 250 ns machine cycle time is successfully developed. Several other features of this processor are also explained and discussed.**

I. Introduction

THE role of the microprocessor has been changing from a replacement of simple logic circuits to the CPU in the main frame. The recent rapid progress in LSI technology has made it possible to realize such complicated microprocessors in general, but some kind of contrivances in circuit design are still necessary to get the higher performance, such as processing speed.

The processing speed is one of the most important performance parameters of the microprocessor, and the improvement of this can be achieved in two ways: by decreasing the executing cycles of each instruction, and by reducing the machine cycle time. These two approaches are related to each other in a given LSI technology and structure, but the reduction of machine cycle time, which is determined by the execution time of the elementary instructions or operations, is more fundamental.

The instructions of a processor can be classified into three categories: 1) data transfer, 2) arithmetic and logic operation, and 3) program or system control. The frequency of the occurrence of the third category is small (less than 20 percent) [1], and the instructions of the first and second categories are generally executed using register-to-register operation (*R-R* operation). For this reason, the signal circulation time through those circuit blocks used in the execution of *R-R* operations substantially determines the machine cycle time.

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Considering these circumstances, we have managed to reduce the signal circulation time of these blocks by introducing the pipelined control methodology, and finally, a new 16 bit microprocessor with 250 ns machine cycle time has been successfully developed.

In this paper, the register file of this microprocessor, a new two-port RAM structure, is described, which plays an important role in realizing the pipelined *R-R* operation. The total structure and some characteristics of this processor are then explained.

II. The Register File and Pipeline Operations

The register file (RF), its peripheral blocks, and the signal flows between those blocks are illustrated in Fig. 1. The execution of an instruction (*R-R* operation, in this case) starts from the instruction register (IR). The instruction bits are divided into the op-code field and the operand field. Each bit of the operand field is fed to the address decoder of RF through a buffer register (BR) where the operand is held during one machine cycle for an adjustment of the timing. On the other hand, each bit of the op-code field is fed to the microprogram controller (MPC) through the interpreting processor (IP); then the microinstructions are decoded to several control signals. One of these signals propagates to the ALU; the other one propagates to the address latches in the RF. It takes about one machine cycle for the control signals to be decoded through the IP, MPC, and DEC. Thus, the control signals and the corresponding address signals reach the RF in the same cycle. If the instruction is *R-R* operation, the gates of these latches turn on; then one register of the RF is accessed. All bits of the operand field are decoded in the RF as an address signal, even if the instruction has nothing to do with the RF.

Functionally, the RF of this processor is a static RAM, having one input/output port to the internal *A*-bus (*A*-port) and one output port to the internal *B*-bus (*B*-port). The static RAM is composed of an address decoder (with dynamic gates to latch the decoded signals), a 16 word × 16 bit matrix of memory cells, and 16 bit I/O cells (which consist of sense amplifiers and the internal bus drivers for the output and a data bit line (DBL) driver for the input).

The timing diagram of the detailed data flow between the RF and ALU is shown in Fig. 2. One machine cycle is divided into four clock phases: *P*1, *P*2, *P*3, and *P*4. The operations in each phase are described as follows.

*P*1: The voltages of address word lines (AWL's) change, and

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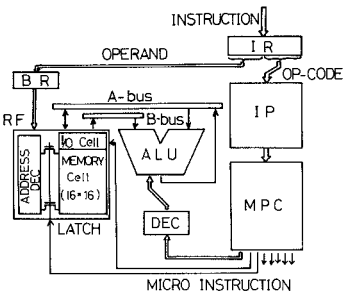

Fig. 1. Block diagram and signal flows of the register file and its peripheral circuits.

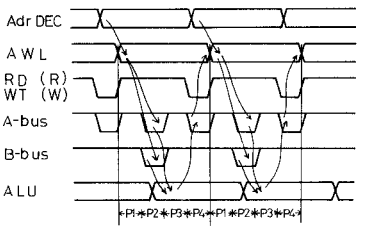

Fig. 2. Timing diagram of register-to-register operation. Address signals decoded in the previous cycle are used for the register access.

a new register is accessed. The data of the accessed register are read out from the memory cells to the I/O cells in the RF block through the DBL's. Two words of data are accessed simultaneously by the A- and B-ports.

$P2$: The data are sent from I/O cells of the RF to ALU through the internal A- and B-buses (in the first half of this phase); then the operation starts at the ALU (in the second half of this phase).

$P3$: The operation completes at the ALU.

$P4$: The data, the result of the operation, are sent back to the RF through the internal A-bus (in the first half of this phase), and are written into the memory cells of the RF (in the second half of this phase). At the same time, the address for the register which will be accessed in the next cycle is decoded in the address decoder of the RF, achieving the pipeline operation.

The internal buses are precharged to the high level at every phase of $P1$ and $P3$.

It is impossible, however, to accomplish the pipeline operation mentioned above by using a conventional RAM structure [2] (which causes a read-time error) when two successive R-R operations are implemented.

A conventional two-port RAM structure, which is a straight extention of the normal static RAM, and the computer-simulated results of this error are illustrated in the solid and dashed line part of Fig. 3 and Fig. 4(a) and (b), respectively. The solid lines of Fig. 4(a) show the input pulses. The accessed memory cell is changed from cell 1 to cell 2, and the mode is changed from write to read. These transitions occur simultaneously at the time point "X" which corresponds to the clock phase turning from $P4$ to $P1$. The corresponding DBL voltages and the state of the cell are shown in Fig. 4(b). Each pair of voltages converges into the same value after 200 ns from "X." This means that the memory cell falls into a conflict state and the read-time error takes place. The cause of

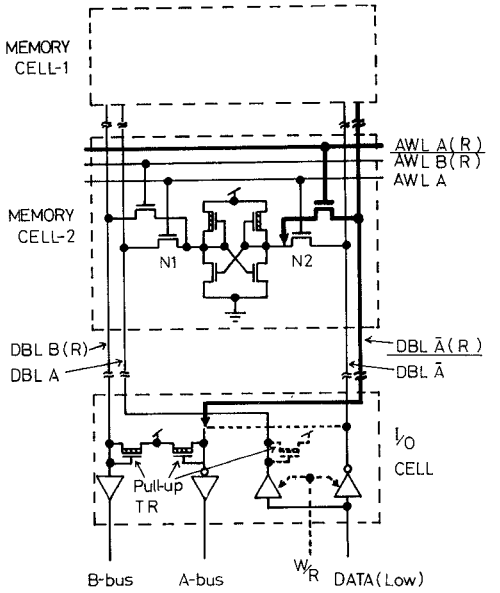

Fig. 3. Circuit diagram of a conventional two-port static RAM (solid and dashed line part), and that of the new register file (solid and bold line part).

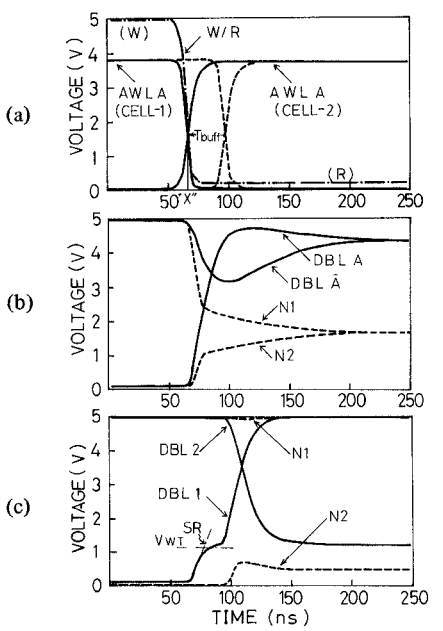

Fig. 4. Results of the computer circuit simulation for a conventional two-port RAM. (a) Input pulses without T_{buff} (solid lines) and with T_{buff} (dashed lines). Voltages of DBL's (solid lines) and the cell nodes (dashed lines) (b) for the inputs without T_{buff} and (c) for the inputs with T_{buff}.

this error is considered to be that the low voltage of the DBL used in the write operation disturbs the memory cell state of the newly accessed register.

The simple way to avoid this error is to provide a buffering clock phase $P41$ between $P4$ and $P1$. The duration of $P41$, T_{buff}, should be longer than T_{min} when the voltage of the DBL would rise more than the write threshold voltage V_{wt}. T_{min} must hold the following relationship:

$$T_{min} = (V_{wt} - V_{dbl|t=wt})/SR$$

where $V_{dbl|t=wt}$ is the voltage of the DBL at the write operation and SR is the slew rate of DBL.A (which is determined by the capacitance of the DBL and the charging current of the pull-up transistor). The dashed lines in Fig. 4(a) show the AWL signal pulses with 30 ns buffering clock phase, and Fig. 4(c) illustrates the corresponding voltages of the DBL and the cell nodes in correct operation. The existence of this clock phase would reduce the processing speed about 25 percent.

III. The Structure of the New RF

It is almost inevitable that two transitions, register address and write-to-read operation, occur simultaneously at the RF in a highly concurrent pipeline-operated processor. To avoid the read-time error without a buffering clock phase, we have adopted a new RF structure. The circuit diagram of the memory cell and the I/O cell of this new RF are illustrated in the solid and bold line part of Fig. 3. The memory cell has two complementary data bit lines, DBL.A and DBL.$\bar{A}$, and one address word line, AWL.A only for a write operation. It also has one data bit line, DBL.$\bar{A}(R)$ or DBL.$B(R)$ and one address-word line, AWL.$A(R)$ or AWL.$B(R)$ only for each read operation of the A- or B-port. In the I/O cell, pull-up transistors and sense amplifiers are connected to the DBL.$\bar{A}(R)$ and DBL.$B(R)$, and the DBL drivers which will drop the DBL voltage below V_{wt} are connected to DBL.A and DBL.$\bar{A}$. In this structure, write and read operations are available almost independently of each other. The voltage of the DBL for the read operation never falls below V_{wt}, so the read operation can be accomplished correctly. A microphotograph of the memory cell of this new RF is shown in Fig. 5. The cell is 61 × 68 μm^2 in area, which is about 30 percent larger than a conventional two-port RAM.

The observation of the new RF operation is carried out by computer circuit simulation and by experimental verification using the test circuit of the new RF with output amplifiers. Fig. 6(a) shows the result of the circuit analysis, and Fig. 6(b) is the oscillograph of the input and output waveforms of the test circuit. The time point "X" is the same critical point, and the output waveform of the sense amplifier changes from low level to high level in less than 20 ns after the time "X." This means that the read operation is correctly accomplished within 20 ns access time.

A memory cell of the two-port RAM has another problem in write operation. Usually, the DBL of the A-port and B-port are connected to the complementary nodes of the memory cell, respectively. When the duration of the write operation (T_{wt}) is short and the supply voltage V_{dd} is low, the operation, writing "0" into the memory cell from the A-port and reading out these data from the B-port in the successive cycle, leads to an erroneous result. Fig. 7 shows the voltage transients of the memory cell in this situation. At the time point "Y" when the memory cell is not yet in the stable state, the read gate of the B-port (AWL.$B(R)$ in Fig. 3) turns on. This introduces the trigger which turns over the memory cell state. The V_{dd} dependency of this write-time error is illustrated in Fig. 8. In the case of the E/E-type AWL driver (shown by the solid line), the minimum duration for a correct write operation ($T_{wt\,\min}$) has its peak at $V_{dd} = 4.5$ V. Two contradictory effects are considered to cause this V_{dd} dependency: 1) the stability of the initial state of the memory cell (the node voltage just before the read operation) increases with V_{dd}, and 2) the trigger energy introduced by the read gate increases with V_{dd}.

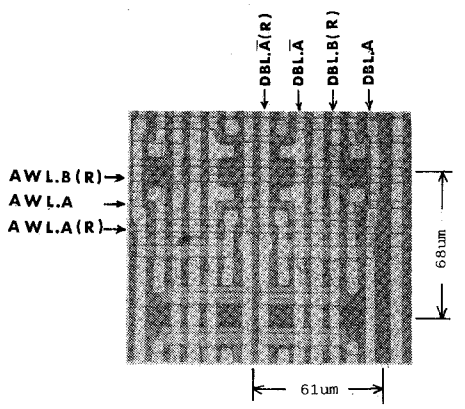

Fig. 5. Microphotograph of the memory cell in the new register file.

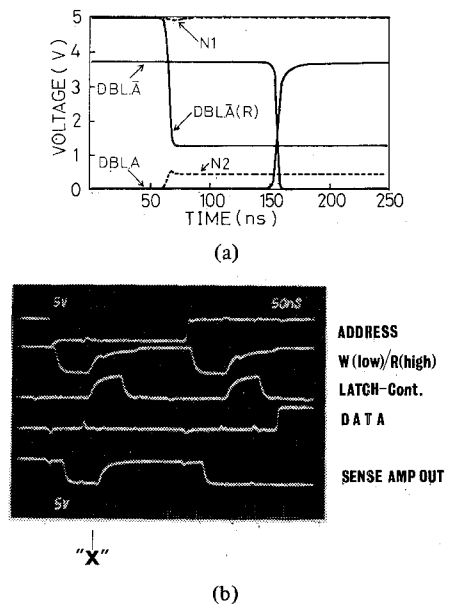

Fig. 6. (a) Results of the computer circuit simulation for the new register file. Voltages of DBL's (solid lines) and cell nodes (dashed lines). (b) Oscillograph of the input and output waveforms for the new register file. "X" is the critical time point.

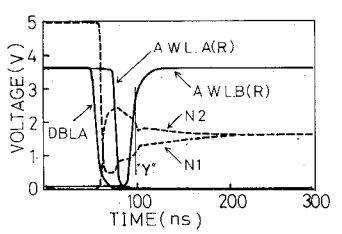

Fig. 7. Results of the computer circuit simulation for the new register file in the situation of write-time error. $V_{dd} = 4.5$ V.

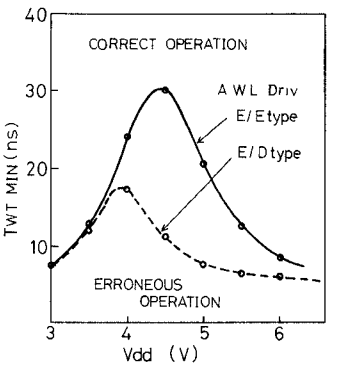

Fig. 8. Measured value of minimum duration for correct write operation as a function of V_{dd}.

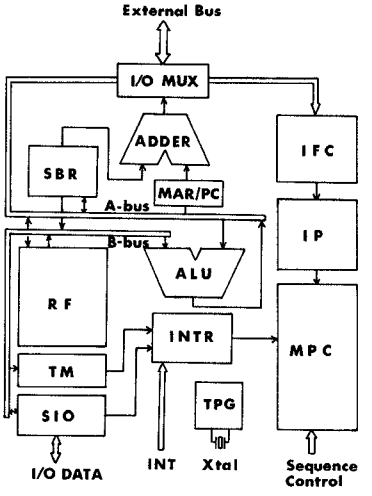

Fig. 9. Block diagram of the 16 bit microprocessor (MN1613).

In the region where V_{dd} is less than 4.5 V, the second effect is dominant; on the other hand, in the higher V_{dd} region, the first effect is dominant. As shown in Fig. 8 (dashed line), by using an E/D-type AWL driver, the write operation becomes faster and $T_{wt\,\min}$ becomes smaller. To avoid this write-time error, we use the E/D-type AWL driver with a little more power dissipation than the E/E-type driver.

IV. 16 Bit Microprocessor

A new 16 bit microprocessor[1] with a 250 ns machine cycle time has been developed. This high speed is achieved mainly by the pipeline operation described in the preceding sections. The chip is primarily designed for control applications. The architecture of this processor is shown in Fig. 9. In addition to the new RF structure, it has other features. These are explained below.

1) For the pipeline operations of instruction fetch, there are two words of prefetch buffers in the instruction fetch controller (IFC). The word size of the buffer has been decided so that the processor has the highest value of "speed/circuit size" when it implements ordinary control programs. The IFC starts to prefetch if any of the prefetch buffers is empty and the I/O multiplexer (I/O MUX) is not occupied by the data I/O operations. Fig. 10 shows the relationship between the average execution time (T_{ex}) of typical control programs and the word size of the buffers (N_{buff}). If the program did not include the branch instruction, T_{ex} would become shorter monotonously with N_{buff}; but in real programs, branch instructions disturb the sequency of instructions, and the prefetched instructions are discarded. In this case, the prefetch operations are not only useless, but also harmful for the execution speed of the processor because they interfere with the data I/O operations inversely. The tradeoff point exists near two words.

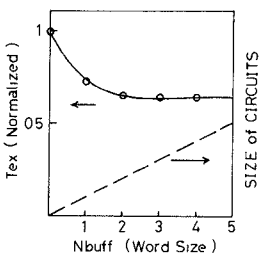

Fig. 10. Measured value of program execution time (T_{ex}) as a function of the word size of prefetch buffer registers (N_{buff}).

[1] MN1613.

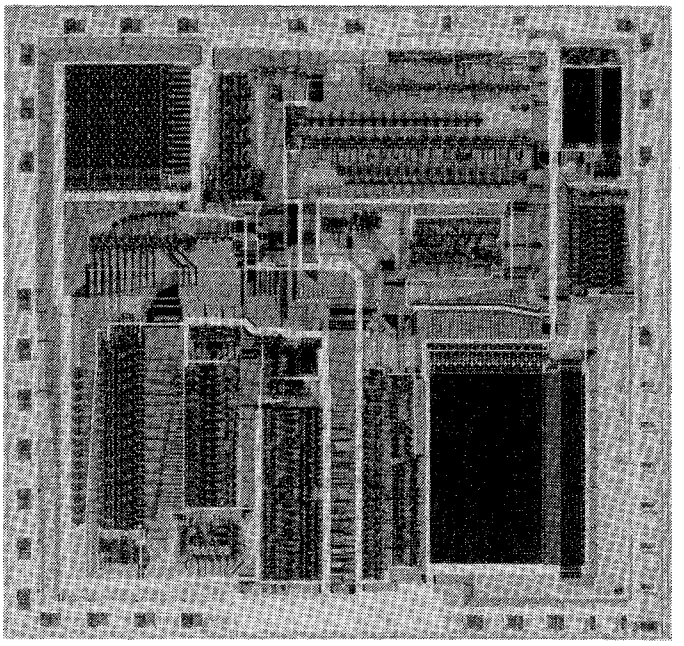

Fig. 11. Microphotograph of the 16 bit microprocessor (MN1613). The chip size is 6.6 × 7.0 mm.

TABLE I
CHARACTERISTICS OF THE LSI

Number of Devices	40 000
Die Size	6.6 × 7.0 mm^2
Process	3 μm NMOS
Power Supply	5 V
Power Dissipation	1.4 W typ.
Package	40*p* DIL
Gate Delay	2.5 ns/gate
I/O Interface	TTL compatible
Machine Cycle Time	250 ns typ.
Instruction Set	97
Address Space	512 kbytes
General Registers	5 words × 16 bits

2) The processor is controlled by the microprogram which is divided into two levels. The first one is in the interpreting processor (IP) (the size is 70 words × 20 bits), and the second one is in the microprogram controller (MPC) (the size is 700 words × 27 bits). The two-level structure is useful to economize the storage size of the control memory by about 20–30 percent and to reduce the coding errors of microinstructions. The microprogramming used in this processor is rather vertical, so further decoding is necessary for many microinstructions from the MPC.

3) A 16 bit timer, an 8 bit serial I/O, and a clock oscillator are included. These functions make the processor suitable for real-time control applications in compact systems.

4) In the operation of addressing, a logical 16 bit address in the memory address register (MAR) or program counter (PC) is added to the 4 bit data in the segment base register (SBR). The joint 18 bit physical address is used for the memory access.

5) The processor has three levels of maskable interrupts which are controlled in the interrupt block (INTR).

The chip is successfully fabricated by Si n-channel *E/D*-MOS technology with a 3 μm design rule. The effective channel length is 2.4 μm and the thickness of the gate oxide is 60 nm.

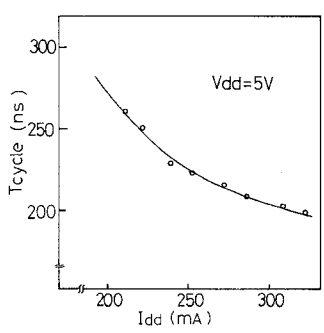

Fig. 12. Measured value of machine cycle time as a function of I_{dd}.

The photomicrograph of the chip is illustrated in Fig. 11. The chip is 6.6 × 7.0 mm^2 in area with 40 000 transistors. Several LSI characteristics are listed in Table I. The machine cycle time is measured as a function of the supply current I_{dd}. The result is illustrated in Fig. 12. This microprocessor has 97

TABLE II
TYPICAL EXECUTION TIMES

Instructions	(μs)
R-R Add/Sub	0.25
16 bit Multiplication	6.25
16 bit Division	13.75
Floating Point Add/Sub	13.0–19.0
Floating Point Multiplication	42.5–49.0
Load/Store (Imd., Index.)	0.75

distinct instructions. The execution times of several instructions are listed in Table II.

V. CONCLUSION

A new register file structure has been developed for a pipeline data operation in a microprocessor. In this structure, read and write operations can be executed in any sequence without errors in the wide range of the supply voltage. Using this register file and the results of the analysis, a 16 bit microprocessor with 250 ns machine cycle time has been designed and successfully fabricated.

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[2] *Memory Design Handbook*, Intel Corp., 3065 Bowers Ave., Santa Clara, CA 95051, 1977, p. 6-2.

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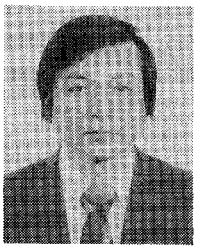

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